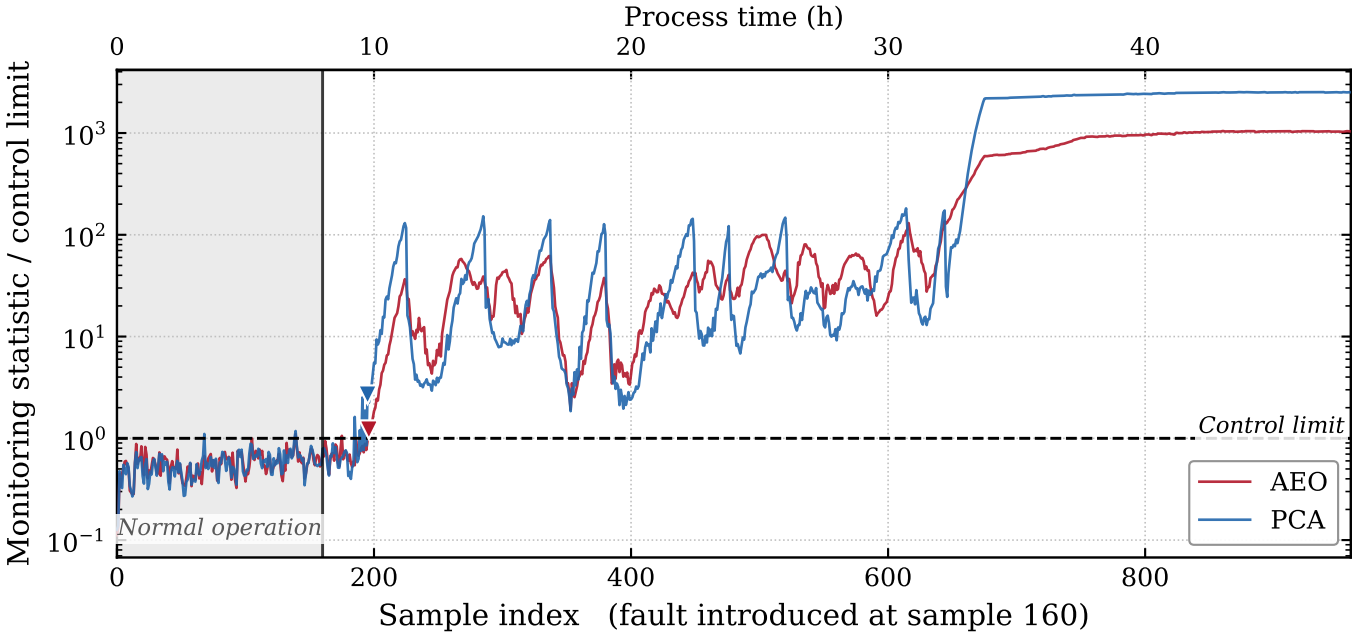
Fault Detection Performance for IDV(18)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	53.7 samples (161 min)	MDR 0.066	detected in 7/7 runs
PCA	delay	49.0 samples (147 min)	MDR 0.059	detected in 7/7 runs

PCA detects 4.7 samples (14 min) earlier; PCA missed-detection rate lower by 11%